

Electromagnetic compatibility — Product family standard for lifts, escalators and moving walks — Immunity

London Underground Ltd

The European Standard EN 12016:2004 has the status of a
British Standard

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National foreword

This British Standard is the official English language version of EN 12016:2004. It supersedes BS EN 12016:1998 which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee MHE/4, Lifts, hoists and escalators, which has the responsibility to:

- aid enquirers to understand the text;
- present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;
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roulants - Immunité

Elektromagnetische Verträglichkeit - Produktfamilien-Norm
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CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CEN member.

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Foreword

This document (EN 12016:2004) has been prepared by Technical Committee CEN/TC 10 “Lifts, escalators and moving walks”, the secretariat of which is held by AFNOR.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by June 2005, and conflicting national standards shall be withdrawn at the latest by June 2006.

This document supersedes EN 12016:1998.

This document has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative Annex ZA, which is an integral part of this document.

The test levels and the performance criteria which are given in this standard reflect the fact that lifts, escalators and moving walks when in use, consist, generally of self contained apparatus (e.g. machine room, car, etc.).

The related EMC product family standard for emission is:

EN 12015:2004, *Electromagnetic compatibility — Product family standard for lifts, escalators and moving walks - Emission*.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

Introduction

This European Standard has been prepared to provide one means of conforming to the requirements of the Electromagnetic Compatibility (EMC) Directive, the Lifts Directive and the Machinery Directive. The requirements of this European Standard have been specified so as to ensure an adequate level of electromagnetic immunity for most cases.

The apparatus concerned and the extent to which hazardous situations and events are covered are indicated in the scope of this document.

Where the provisions of this type C standard are different from those which are stated in type A or B standards, the provisions of this type C standard take precedence over the provisions of the other standards, for apparatus that have been designed and built according to the provisions of this type C standard.

Test levels and immunity performance criteria are defined for:

- apparatus which are safety components or are used in conjunction with safety components, (safety circuits) ;
- apparatus used in general function circuits.

The test levels and requirements are given on the basis that the apparatus, generally, is connected to a low voltage system.

The requirements for safety circuits provide one means of demonstrating conformity with the essential health and safety requirements of the Lifts Directive and the Machinery Directive with regard to EMC immunity.

Due to the size of an installed lift, it becomes impracticable to test the total assembly either in a test laboratory or in situ where the uncontrolled environment may also influence the test procedures and results. This applies also to measurements within the car. Similar considerations regarding dimensions apply equally to the testing of escalators and moving walks.

The following explains the rationale to the revision of the standard EN 12016:1998:

a) Important changes

The scope excludes severe electromagnetic environments and apparatus already proven to be in conformity with the Electromagnetic Compatibility Directive.

The term "installation" has been changed to "system". This is due to the fact that official interpretation defines that fixed installations are not covered by the conformity assessment requirements of the EMC Directive. The scope of the standard is applicable to the apparatus and assembly of apparatus of lifts and escalators and assembly into systems.

New requirements on radio frequency electromagnetic field above 500 MHz, these are extended to cover the digital mobile telephone services up to 1960 MHz.

New requirements for surge testing on safety circuits.

New requirements on Radio frequency electromagnetic field regarding safety devices as defined by the Lifts Directive and mobile telephones or radio transmitters as a result of a risk assessment. It is assumed that mobile telephones and radio transmitters are not used at frequencies up to 166 MHz near safety circuits of equipment covered by the scope of this standard.

Higher requirements on several environmental phenomena considering the progress on EMC technology and the results of the risk assessment.

New requirements have been introduced for immunity to mains power supply voltage interruptions and voltage dips.

b) Environmental issues

Lifts, escalators and moving walks are systems whose component apparatus/assembly of apparatus are distributed (and some of which move) throughout the building. The definition in EMC terms of the use of the building (residential or industrial) cannot be predetermined or assumed to be fixed. Therefore, to cover requirements in all cases, no differentiation between environments has been made and a single set of limits has been maintained.

Severe electromagnetic environments have not been considered. Examples of these are: radio transmitter stations, railways and metros, heavy industrial plant, electricity power stations. Additional tests and immunity measures may need to be taken on apparatus to be used in these environments.

It is assumed that no ports connected to safety circuit only are rated at currents greater than 100 amps.

1 Scope

This European Standard specifies the immunity performance criteria and test levels for apparatus used in lifts, escalators and moving walks which are intended to be permanently installed in buildings including the basic safety requirements in regard to their EMC environment. These levels represent essential EMC requirements.

The standard refers to normal EMC conditions as existing in residential, office and industrial buildings, but does not cover more severe EMC environments such as:

- radio transmitter stations;
- railways and metros;
- heavy industrial plant;
- electricity power stations

which need additional investigations.

This standard addresses commonly known EMC related hazards and hazardous situations relevant to lifts, escalators and moving walks when they are used as intended and under the conditions foreseen by the lift installer or escalator and/or moving walk manufacturer.

This standard addresses the environmental conditions stated in the EN 81 series of standards and EN 115 (humidity, temperature, etc.), so far as they are related to EMC performance.

However:

- performance criteria and test levels for apparatus/assembly of apparatus used in general function circuits do not cover situations with an extremely low probability of occurrence;
- this standard does not apply to other apparatus already proven to be in conformity to the EMC Directive, and not related to the safety of the lift, escalator or moving walk, such as lighting apparatus, communication apparatus, etc.

This standard is not applicable to lifts, escalators, moving walks and related safety components manufactured before the date of publication of this standard by CEN.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 81-1, *Safety rules for the construction and installation of lifts — Part 1: Electric lifts.*

EN 81-2, *Safety rules for the construction and installation of lifts — Part 2: Hydraulic lifts.*

EN 115, *Safety rules for the construction and installation of escalators and passenger conveyors.*

EN 1070, *Safety of machinery – Terminology.*

EN 61000-4-2, *Electromagnetic Compatibility (EMC) — Part 4-2: Testing and measurement techniques; Electrostatic discharge immunity test (IEC 61000-4-2:1995).*

EN 61000-4-3, *Electromagnetic compatibility (EMC) — Part 4-3: Testing and measurement techniques; Radiated, radio-frequency, electromagnetic field immunity test (IEC 61000-4-3:2002).*

EN 61000-4-4, *Electromagnetic compatibility (EMC) — Part 4-4: Testing and measurement techniques — Electrical fast transient/burst immunity test* (IEC 61000-4-4:1995).

EN 61000-4-5, *Electromagnetic compatibility (EMC) — Part 4-5: Testing and measurement techniques; Surge immunity test* (IEC 61000-4-5:1995).

EN 61000-4-6, *Electromagnetic Compatibility (EMC) — Part 4-6: Testing and measurement techniques; Immunity to conducted disturbances, induced by radio-frequency fields* (IEC 61000-4-6:1996).

EN 61000-4-11, *Electromagnetic Compatibility (EMC) — Part 4-11: Testing and measurement techniques — Voltage dips, short interruptions and voltage variations immunity tests* (IEC 61000-4-11:1994).

EN 61000-6-1, *Electromagnetic compatibility – Part 6-1: Generic standards – Immunity for residential, commercial and light-industrial environments* (IEC 61000-6-1:1997, modified).

EN 61000-6-2, *Electromagnetic Compatibility (EMC) — Part 6-2: Generic standards - Immunity for industrial environments* (IEC 61000-6-2:1999, modified).

IEC 60050-161, *International Electrotechnical Vocabulary - Chapter 161: Electromagnetic compatibility*.

3 Terms and definitions

For the purpose of this European Standard, the terms and definitions given in EN 1070, EN 61000-6-1, EN 61000-6-2, IEC 60050-161 and the following apply.

3.1

system

lift escalator or moving walk comprising assembly of apparatus with electrical and electronic equipment and interconnections

NOTE See Figure 1 and Figure 2 as examples.

3.2

assembly of apparatus

arrangement of interconnected apparatus, which can be tested together

NOTE See Figure 1 and Figure 2 as examples.

3.3

apparatus

assembly of components with an intrinsic function as defined by its manufacturer

NOTE 1 See Figure 1 and Figure 2 as examples.

NOTE 2 Safety components listed by Annex IV of the Lifts Directive are considered as apparatus.

3.4

port

particular interface of specified apparatus/assembly of apparatus with the external electromagnetic environment

NOTE See Figure 3 as example.

3.5

enclosure port

physical boundary of apparatus/assembly of apparatus through which electromagnetic fields may radiate or impinge

NOTE See Figure 3 as example.

3.6

safety circuit

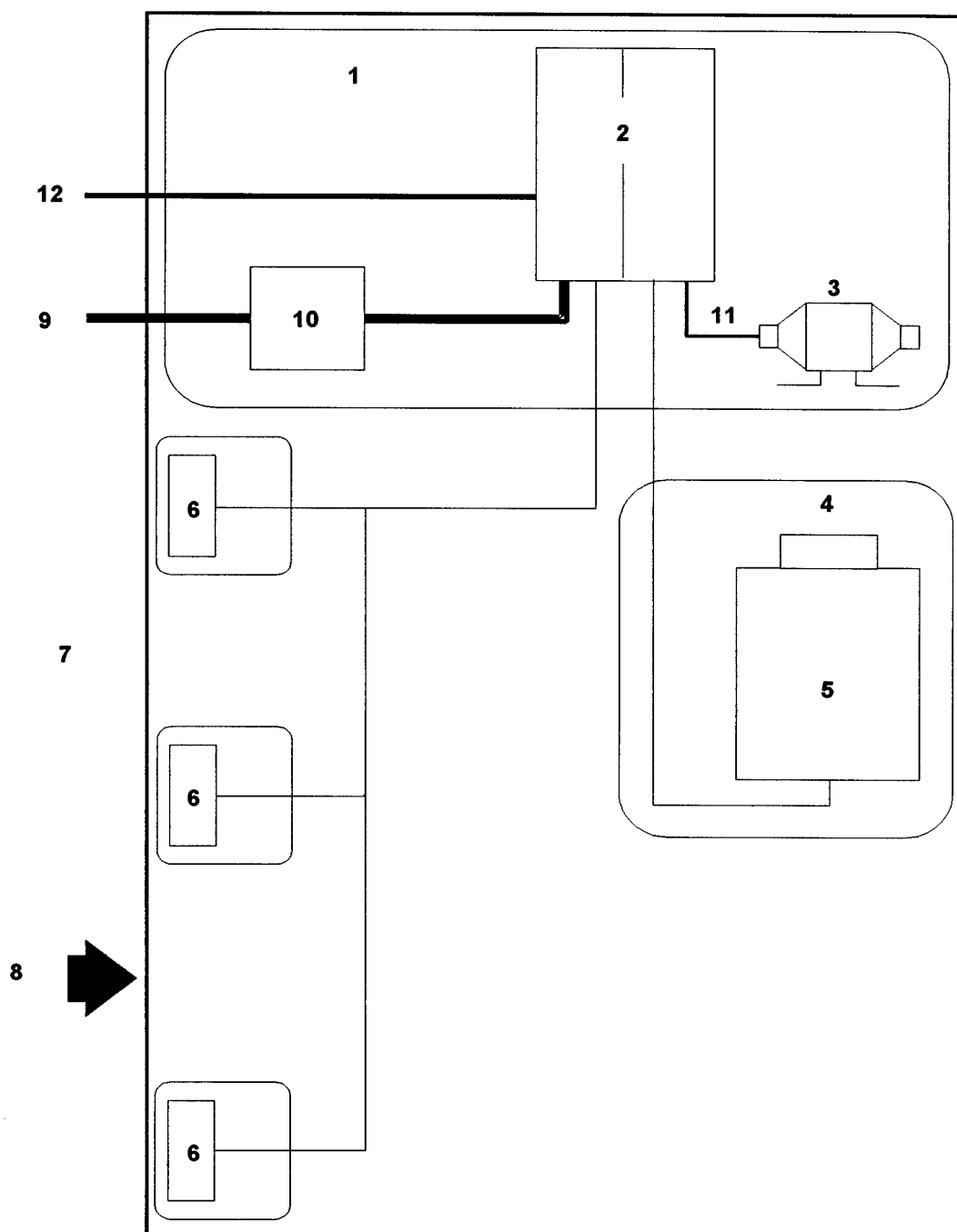
circuit containing electronic components forming an electric safety device as defined in EN 81-1, EN 81-2

NOTE Safety components listed by Annex IV of the Lifts Directive are considered to be safety circuits.

3.7

general function circuit

circuitry used in apparatus which does not incorporate safety circuits

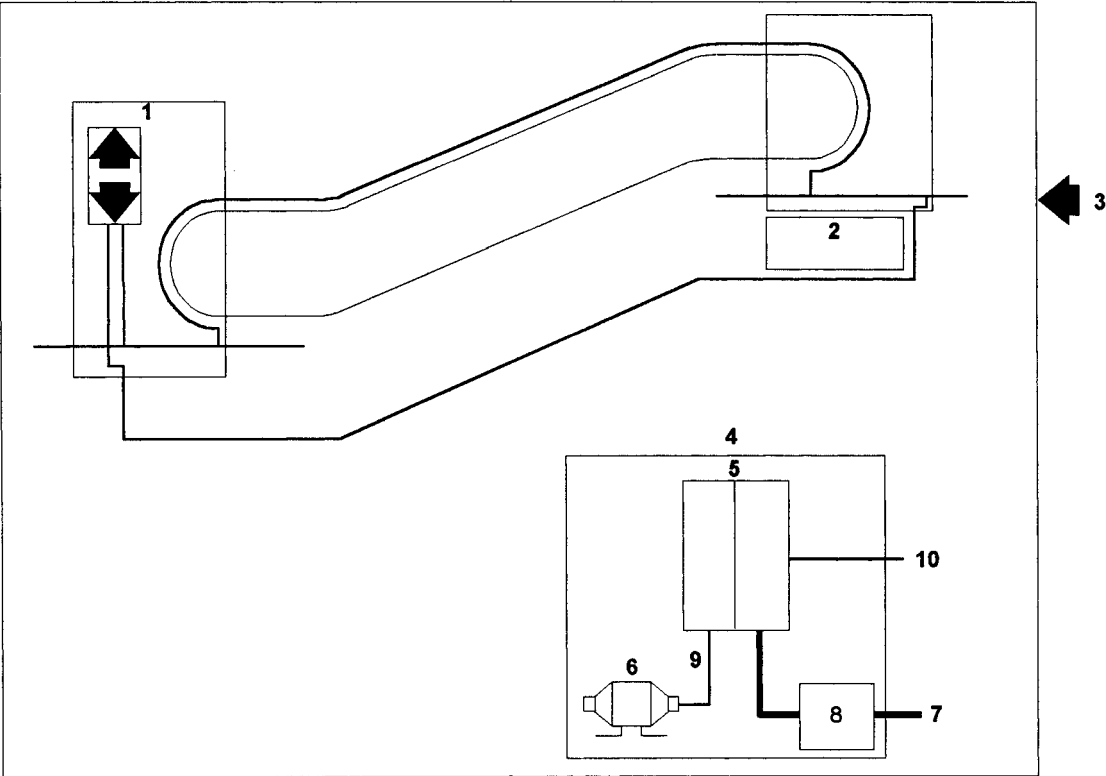


Key



1	Machinery space	7	Landings
2	Main control / control cabinet	8	System boundary
3	Machine	9	AC – and/or DC power ports
4	Door control	10	Main switch
5	Lift car	11	Output power port
6	Apparatus installed at the landing (e.g. push buttons, indicators)	12	Ports for monitoring and remote alarm systems (Signal and control ports)

Figure 1 — EMC example (immunity) for lift systems



Key



- | | | | |
|------|---|----|---|
| 1 | Control panels | 6 | Machine |
| 2 | Machinery space (see 4 to 10) | 7 | AC – and/or DC power ports |
| NOTE | The machinery space can also be an external room. | | |
| 3 | System boundary | 8 | Main switch |
| 4 | Machinery space | 9 | Output power port |
| 5 | Main control/control cabinet | 10 | Ports for monitoring (Signal and control ports) |

Figure 2 — EMC example (immunity) for escalator and moving walk systems

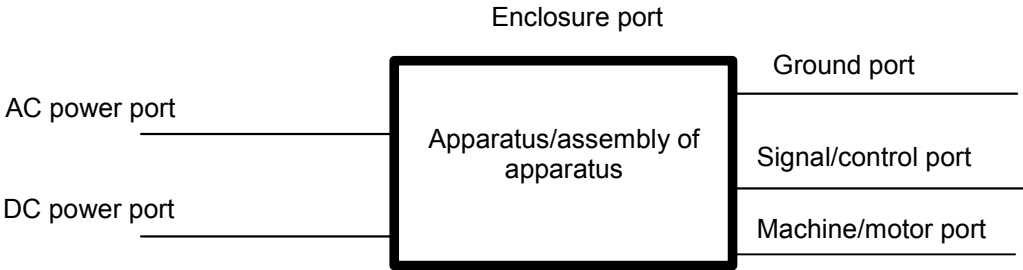


Figure 3 — Examples of ports

4 Test procedure

4.1 The tests shall be carried out as stated in EN 61000-4-2, EN 61000-4-3, EN 61000-4-4, EN 61000-4-5, EN 61000-4-6, EN 61000-4-11, as appropriate (see Tables 1 to 7).

An attempt shall be made to maximise the susceptibility by varying the position of the test sample in the test facility.

4.2 Travelling cables or any other cables likely to be more than 5 m long shall be represented by a sample of at least 5m long connected to the relevant port for the purpose of testing for susceptibility.

4.3 It is not always possible to measure the immunity levels for every function of the apparatus/assembly of apparatus. In such cases the most critical period of operation shall be selected.

4.4 The test shall be carried out at a single set of environmental conditions within the specified operating range of temperature, humidity, air pressure and supply voltage, unless otherwise indicated in the standards referred to in 4.1.

4.5 Tests shall be carried out in well-defined and reproducible conditions for each test.

4.6 To facilitate testing, assemblies of apparatus can be tested together. However where apparatus contain safety circuits, the tests shall prove that the safety circuits comply with the immunity requirements for all circuits and specific requirements for safety circuits.

This does not imply that those parts of the whole assembly, which are general function circuits, have to comply with the safety circuit requirements.

4.7 Tests shall be carried out at the following ports of the apparatus or assembly of apparatus where they exist:

- enclosure port (see Table 1);
- ports for signal and control lines (see Table 2);
- ports for monitoring and remote alarm systems crossing the system boundaries (see Table 3)
- input and output D.C. power ports with current rating ≤ 100 A (see Table 4);
- input and output D.C. power ports with current rating > 100 A (see Table 5);
- input and output A.C. power ports with current rating at ≤ 100 A per phase (see Table 6) ;
- input and output A.C. power ports with current rating > 100 A per phase (see Table 7).

4.8 The configuration and mode of operation during measurement shall be recorded.

4.9 Test values shall be applied as specified in Tables 1 to 7, and the performance criteria in 6.2 shall be met as appropriate. The tests shall be carried out individually as single tests and in sequence.

5 Applicability of tests

The application of tests for evaluation of immunity depends on the type of apparatus/assembly of apparatus, its configuration, ports, technology and operating conditions.

5.1 It might be determined from consideration of the electrical characteristics and usage of a particular apparatus/assembly of apparatus that some of the tests are inappropriate and therefore unnecessary. In such a case the decision and justification not to test shall be recorded.

5.2 Where deviations from the test methods specified in **4.1** are applied, such deviations shall be justified and recorded.

6 Evaluation of tests results

6.1 Introduction

The special requirements of the product family make it necessary to define precise criteria for the evaluation of the immunity test results.

As stated in the scope of this standard, the basic safety requirements of the system (lift, escalator or moving walk) are specified in EN 81-1, EN 81-2 and EN 115. For this reason a performance criterion derived from the immunity levels of EN 61000-6-1 is deemed sufficient for most functions. However, for safety circuits, where any malfunction that may produce an unsafe operating mode cannot be tolerated, higher immunity levels derived from EN 61000-6-2 are stipulated.

6.2 Performance criteria

A functional description of the apparatus or assembly of apparatus and a definition of performance criteria, during or as a consequence of testing shall be recorded.

The definition of performance criteria shall be based on:

Performance criterion A: The apparatus/assembly of apparatus shall continue to operate as intended. No degradation of performance or loss of function is allowed below a performance level specified by its manufacturer, when the apparatus or assembly of apparatus is used as intended. In some cases the performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer then either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus or assembly of apparatus if used as intended.

Performance criterion B: The apparatus/assembly of apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by its manufacturer, when the apparatus or assembly of apparatus is used as intended. In some cases the performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is however allowed. No change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer then either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus or assembly of apparatus if used as intended.

Performance criterion C: Not used.

Performance criterion D: The apparatus/assembly of apparatus and the associated safety component(s) shall continue to operate as intended. No degradation of performance or loss of function is allowed other than a failure into a safe mode.

6.3 Enclosure ports of safety circuits

Where a risk assessment demonstrates that the location of a safety circuit may allow a mobile telephone or emergency services radio transmitter to be placed directly beside or on the device, then the enclosure port of the device shall be implemented so that the performance criterion D is satisfied in these conditions.

7 Documentation for the installer of the apparatus/assembly of apparatus

The installer shall be provided with documentary information and instructions for installation and use so that compliance with this standard is maintained.

This shall include where applicable:

- instructions for assembly and physical arrangement with other apparatus;
- instructions and precautions for interconnection to other apparatus;
- specifications of interconnection cables and devices;
- instructions for commissioning and testing;
- guidance on avoiding incorrect actions and assembly of apparatus which are known to cause non-compliance with this standard.

EN 12016:2004 (E)

Table 1 — Immunity - Enclosure port

Environmental phenomena	Test set-up	Units	Test values		Performance criteria	
			All circuits ^a	Safety circuits ^b	All circuits ^a	Safety circuits ^b
Electrostatic discharge ^e	EN 61000-4-2	kV (charge voltage)	4 Contact 8 Air discharge	6 Contact 15 Air discharge	B	D
Radio frequency electromagnetic field ^c	EN 61000-4-3	MHz V/m (rms, unmodulated) % AM (1 kHz)	80 to 166 10 80	80 to 166 10 ^d 80	A	D
Radio frequency electromagnetic field ^c	EN 61000-4-3	MHz V/m (rms, unmodulated) % AM (1 kHz)	166 to 1000 10 80	166 to 1000 30 ^d 80	A	D
Radio frequency electromagnetic field ^c	EN 61000-4-3	MHz V/m (rms, unmodulated) % AM (1 kHz)	1710 to 1784 10 80	1710 to 1784 30 ^d 80	A	D
Radio frequency electromagnetic field ^c	EN 61000-4-3	MHz V/m (rms, unmodulated) % AM (1 kHz)	1880 to 1960 3 80	1880 to 1960 10 ^d 80	A	D
^a Test values for ports containing general function and/or safety circuits. ^b Test values for ports containing safety circuits. ^c For the relationship between test levels, protection distances and radiated powers of mobile telephones, EN 61000-4-3. ^d Field strength may be limited by distance on site, e.g. 200 mm for mobile phones; therefore if no RF shielding barrier is used then a physical barrier shall maintain a distance of 200 mm between the safety circuit and potential sources of perturbation. ^e If safety circuits are not in a grounded metal box, at least 8 mm of distance between box and circuitry shall be applied to avoid on site damage or other type of insulation shall be used.						

Table 2 — Immunity - Ports for signal and control lines

Environmental phenomena	Test set-up	Units	Test values		Performance criteria	
			All circuits ^a	Safety circuits ^b	All circuits ^a	Safety circuits ^{b)}
Fast transients common mode	EN 61000-4-4	kV (peak) T _r /T _h (ns) Rep. Frequency (kHz)	0,5 5/50 5	2,0 5/50 5	B	D
Surge Line to ground Line to line	EN 61000-4-5	T _r /T _h (μs) kV (peak) kV (peak)	Not relevant	1.2/50 +/- 2,0 +/- 1,0	Not relevant	D
Radio frequency common mode	EN 61000-4-6	MHz V (rms, unmodulated) % AM (1 kHz)	0,15 to 80 3 80 ^{c, d}	0,15 to 80 10 80 ^d	A	D
NOTE T _r is the rise time of the pulse, T _h is the time duration of the pulse, shape of the pulse is defined in the relevant test setup standard.						
^a Test values for ports containing general function and/or safety circuits. ^b Test values for ports containing safety circuits. ^c Applicable only to ports interfacing with cables whose total length according to the manufacturers functional specification may exceed 3m. ^d The test level can also be defined as the equivalent current into a 150 Ω load.						

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Table 3 — Immunity - Ports for monitoring and remote alarm systems crossing the system boundaries not applicable to input ports intended for connection to dedicated non-rechargeable power supplies

Environmental phenomena	Test set-up	Units	Test values		Performance criteria	
			All circuits ^a	Safety circuits ^b	All circuits ^a	Safety circuits ^b
Fast transients common mode	EN 61000-4-4	kV (peak) T _r /T _h (ns) Rep. Frequency (kHz)	+/- 1,0 5/50 5 ^c	Covered by Table 2	B	Refer to Table 2
Surge Line to ground Line to line	EN 61000-4-5	T _r /T _h (μs) kV (peak) kV (peak)	1,2/50 +/- 1,0 +/- 0,5	Covered by Table 2	B	Refer to Table 2
Radio frequency common mode	EN 61000-4-6	MHz V (rms, unmodulated) % AM (1 kHz)	0,15 - 80 3 80 ^{c, d}	Not relevant	A	Not relevant
NOTE T _r is the rise time of the pulse, T _h is the time duration of the pulse, shape of the pulse is defined in the relevant test setup standard.						
^a Test values for ports containing general function and/or safety circuits. ^b Test values for ports containing safety circuits. ^c Applicable only to ports interfacing with cables whose total length according to the manufacturers functional specification may exceed 3m. ^d The test level can also be defined as the equivalent current into a 150 Ω load.						

Table 4 — Immunity - Input and output D.C. power ports with current rating ≤ 100 A not applicable to input ports intended for connection to dedicated non-rechargeable power supplies

Environmental phenomena	Test set-up	Units	Test values		Performance criteria	
			All circuits ^a	Safety circuits ^b	All circuits ^a	Safety circuits ^b
Fast transients common mode	EN 61000-4-4	kV (peak)	0,5	4,0	B	D
		T _r /T _h (ns)	5/50	5/50		
		Rep. Frequency (kHz)	5 ^d	2,5		
Surge	EN 61000-4-5	T _r /T _h (μs)	1,2/50	1.2/50	B	D
Line to ground		kV (peak)	+/- 0,5	+/- 2,5		
Line to line		kV (peak)	+/- 0,5 ^d	+/- 1,0		
Radio frequency common mode	EN 61000-4-6	MHz	0,15 - 80	0,15 - 80	A	D
		V (rms, unmodulated)	3	10		
		% AM (1 kHz)	80 ^c	80 ^c		
NOTE T _r is the rise time of the pulse, T _h is the time duration of the pulse, shape of the pulse is defined in the relevant test setup standard.						
^a Test values for ports containing general function and/or safety circuits.						
^b Test values for ports containing safety circuits.						
^c The test level can also be defined as the equivalent current into a 150 Ω load.						
^d Applicable only to input ports crossing the system boundary.						

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Table 5 — Immunity - Input and output D.C. power ports with current rating > 100 A not applicable to input ports intended for connection to dedicated non-rechargeable power supplies

Environmental phenomena	Test set-up	Units	Test values		Performance criteria	
			All circuits ^a	Safety circuits ^b	All circuits ^a	Safety circuits ^b
Fast transients common mode	EN 61000-4-4	kV (peak) T _r /T _h (ns) Rep. Frequency (kHz)	1,0 5/50 5 ^d	Not relevant	B	Not relevant
Surge Line to ground Line to line	EN 61000-4-5	T _r /T _h (μs) kV (peak) kV (peak)	1,2/50 +/- 1,0 +/- 0,5 ^d	Not relevant	B	Not relevant
Radio frequency common mode	EN 61000-4-6	MHz V (rms, unmodulated) % AM (1 kHz)	0,15 – 80 3 80 ^c	Not relevant	A	Not relevant
NOTE T _r is the rise time of the pulse, T _h is the time duration of the pulse, shape of the pulse is defined in the relevant test setup standard.						
^a Test values for ports containing general function and/or safety circuits.						
^b Test values for ports containing safety circuits.						
^c The test level can also be defined as the equivalent current into a 150 Ω load.						
^d Applicable only to input ports crossing the system boundary.						

Table 6 — Immunity - Input and output A.C. power ports rated at ≤ 100 A per phase not applicable to input ports intended for connection to dedicated non-rechargeable power supplies

Environmental phenomena	Test set-up	Units	Test values		Performance criteria	
			All circuits ^a	Safety circuits ^b	All circuits ^a	Safety circuits ^b
Fast transients common mode	EN 61000-4-4	kV (peak) T_r/T_h (ns) Rep. Frequency (kHz)	1,0 5/50 5	4.0 5/50 2,5	B	D
Voltage dips	EN 61000-4-11	% reduction ms	30 10	30 and 60 ^e 10 to 100 in 10 ms steps 200 to 1000 in 100 ms steps	A	D
Voltage interruptions	EN 61000-4-11 ^c	% reduction ms	Not relevant	>95 ^e 5000	Not relevant	D
Surge Line to ground Line to line	EN 61000-4-5	T_r/T_h (μ s) kV (peak) kV (peak)	1,2/50 +/- 2,0 +/- 1,0 ^f	1,2/50 +/- 2,0 +/- 1,0	B	D
Radio frequency common mode	EN 61000-4-6	MHz V (rms, unmodulated) % AM (1 kHz)	0,15 to 80 3 80 ^d	0,15 to 80 10 80 ^d	A	D

NOTE T_r is the rise time of the pulse, T_h is the time duration of the pulse, shape of the pulse is defined in the relevant test setup standard.

^a Test values for ports containing general function and/or safety circuits.

^b Test values for ports containing safety circuits.

^c See also EN 61800-3:1996, 5.2.2.

^d The test level can also be defined as the equivalent current into a 150 Ω load.

^e The safety circuit shall go into a safe state if the voltage falls below the designed functional limits.

^f Only applicable to input ports.

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Table 7 — Immunity - Input and output A.C. power ports rated at >100 A per phase

Environmental phenomena	Test set-up	Units	Test values		Performance criteria	
			General function circuits ^a	Safety circuits ^b	General function circuits ^a	Safety circuits ^b
Fast transients common mode	EN 61000-4-4	kV (peak) T _r /T _h (ns) Rep. Frequency (kHz)	+/- 2,0 5/50 5	Not relevant	B	Not relevant
Surge ^c Line to ground Line to line	EN 61000-4-5	T _r /T _h (μs) kV (peak) kV (peak)	1,2/50 +/- 2,0 +/- 1,0	Not relevant	B	Not relevant
Radio frequency common mode	EN 61000-4-6	MHz V (rms, unmodulated) % AM (1 kHz)	0,15 - 80 3 80 ^d	Not relevant	A	Not relevant

NOTE T_r is the rise time of the pulse, T_h is the time duration of the pulse, shape of the pulse is defined in the relevant test setup standard.

^a Test values for ports containing general function circuits only.

^b Test values for ports containing safety circuits.

^c Only applicable to input AC ports.

^d The test level can also be defined as the equivalent current into a 150 Ω load.

Annex ZA (informative)

Relationship between this European Standard and the Essential Requirements of EU Directives

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association to provide one means of conforming to Essential Requirements of the following EU Directives:

- EMC Directive 89/336/EEC, as amended by Directives 91/263/EEC, 92/31/EEC, 93/68/EEC and 93/97/EEC;
- Machinery Directive 98/37/EC, amended by Directive 98/79/EC;
- Lifts Directive 95/16/EC.

Once this standard is cited in the Official Journal of the European Communities under that Directive and has been implemented as a national standard in at least one Member State, compliance with the normative clauses of this standard confers, within the limits of the scope of this standard, a presumption of conformity with the relevant Essential Requirements 1.5.11 of the Machinery Directive and associated EFTA regulations.

WARNING — Other requirements and other EU Directives may be applicable to the product(s) falling within the scope of this standard.

Bibliography

- [1] EN 61800-3:1996, *Adjustable speed electric power drive systems - Part 3: EMC product standard including specific test methods (IEC 61800-3:1996)*.

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